

The empirical recurrence for OEIS A279131, recovered from its tail

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Abstract

OEIS A279131 records “Empirical recurrence of order 88” and keeps the recurrence itself in a linked file rather than in the entry, so the conjecture cannot be read off the entry at all. It can still be settled. The entry counts arrays under a condition local to a bounded window of lines, which makes the count a walk count on a finite digraph and forces a monic integer recurrence of order at most the number $S' = 1550$ of states with distinct futures. Berlekamp–Massey on enough exact terms therefore returns the minimal such recurrence, and on the tail beginning at $n = 21$, so the recurrence is proved for every $n \geq 109$. Its last coefficient is $c_{88} = -23393656 \neq 0$, so no further tail satisfies anything shorter; any order-88 recurrence the sequence satisfies from any point on must then have the same characteristic polynomial. The entry’s recurrence is the one displayed here, whatever its linked file contains, and it is proved.

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1 A conjecture that is not written down

OEIS A279131 is “Number of $n \times 5$ 0..1 arrays with no element equal to a strict majority of its horizontal and vertical neighbors, with the exception of exactly two elements, and with new values introduced in order 0 sequentially upwards.”. It has offset 1 and begins

3, 87, 820, 6396, 47452, 316516, 2017028, 12376570, ...

The entry states:

Empirical recurrence of order 88 (see link above)

As of the “Last modified” line on the live entry (Dec 06 2016 19:01:34, revision 6) this is still recorded as empirical, and it is still open. The coefficients are nowhere in the entry text: what is conjectured is only that a recurrence of that order exists.

2 The count is a walk count

The entry’s condition is local. It constrains a bounded window of consecutive lines of the array and nothing beyond it, so the admissible configurations of one window are the vertices of a finite digraph, an edge joins u to v when v may follow u , and an admissible array is exactly a walk. Writing M for the adjacency matrix and ι, τ for the indicator vectors of the admissible first and last windows,

$$a(n) = \iota^\top M^{j(n)} \tau$$

for a linear reindexing j fixed by the entry’s shape. States with identical futures contribute identically to that product and may be merged without changing any count; after merging $S' = 1550$ states remain, and it is that bound every computation below uses.

Lemma 1. *a satisfies a monic linear recurrence with integer coefficients of order at most $S' = 1550$.*

Proof. By the Cayley–Hamilton theorem $M^{S'}$ is an integer combination of $I, M, \dots, M^{S'-1}$, and multiplying that relation on the left by $\iota^\top M^j$ and on the right by τ turns it into a relation among $S' + 1$ consecutive values of a . $\square$

3 Why the question has to be asked of a tail

Running Berlekamp–Massey on the sequence from its first term returns an order LARGER than 88: the opening terms do not satisfy the recurrence, and a recurrence that holds only eventually always looks longer when the exceptional terms are included. That is not evidence against the entry. An OEIS empirical recurrence is asserted for n beyond some index, not from the first term, so the question has to be asked of the tails. It was asked of each tail in turn, and the tail beginning at $n = 21$ is the first whose minimal order is exactly the stated 88.

Lemma 2. *A sequence known to satisfy some linear recurrence of order at most S' is determined, together with its minimal recurrence, by any $2S'$ consecutive terms: Berlekamp–Massey applied to them returns that minimal recurrence exactly.*

Applied to $2S' = 3100$ exact terms from $n = 21$ on, it returns order 88 — the order the entry states — with integer coefficients:

$$a(n) = \sum_{i=1}^{88} c_i a(n-i),$$

c_1	28	c_{23}	10279951659	c_{45}	−871960593031	c_{67}	3779260883784
c_2	−348	c_{24}	12082413515	c_{46}	112022492932	c_{68}	1584273792717
c_3	2526	c_{25}	−41644321745	c_{47}	1154327827131	c_{69}	−240167851692
c_4	−11706	c_{26}	21308485083	c_{48}	204163266121	c_{70}	−1336525087749
c_5	34320	c_{27}	73392938309	c_{49}	−958780400173	c_{71}	−1490049985611
c_6	−50717	c_{28}	−136942235570	c_{50}	946387687653	c_{72}	−636439664438
c_7	−41032	c_{29}	7451049918	c_{51}	686991719669	c_{73}	5547781289
c_8	360939	c_{30}	261976701772	c_{52}	−2414448972083	c_{74}	325143177189
c_9	−519915	c_{31}	−309923342965	c_{53}	−1205865567540	c_{75}	304628047263
c_{10}	−1067628	c_{32}	−98942593605	c_{54}	−427547198844	c_{76}	142994161011
c_{11}	5381820	c_{33}	553180479700	c_{55}	−404562771621	c_{77}	21116275152
c_{12}	−5537875	c_{34}	−346068174427	c_{56}	−18920768844	c_{78}	−44089423750
c_{13}	−16738115	c_{35}	−459567067866	c_{57}	2082302257895	c_{79}	−39174327956
c_{14}	61324131	c_{36}	796664850586	c_{58}	4095289952554	c_{80}	−17883684024
c_{15}	−45450058	c_{37}	7188196958	c_{59}	2880653321001	c_{81}	−2855183590
c_{16}	−164598392	c_{38}	−973565493267	c_{60}	449225174628	c_{82}	2979728581
c_{17}	439100484	c_{39}	739402088571	c_{61}	−1882976582196	c_{83}	2592107445
c_{18}	−64665322	c_{40}	431509482654	c_{62}	−3543427737711	c_{84}	1252797271
c_{19}	−1508859326	c_{41}	−826326303642	c_{63}	−5223763067538	c_{85}	168158069
c_{20}	2543906343	c_{42}	−98383120282	c_{64}	−2958788880573	c_{86}	−128719734
c_{21}	1155468725	c_{43}	504204580207	c_{65}	1148305470525	c_{87}	−82859348
c_{22}	−9607043303	c_{44}	187286968470	c_{66}	3481480112211	c_{88}	−23393656

4 The recurrence, and the identification

Theorem 3. *The recurrence above holds for every $n \geq 109$, and it is the recurrence OEIS A279131 records.*

Proof. That it holds is Lemma 2 applied to the tail from $n = 21$, whose minimal recurrence it is.

For the identification, write $q_{\min}$ for the characteristic polynomial of that minimal recurrence, $x^{88} - c_1x^{88-1} - \dots - c_{88}$. Its constant term is $-c_{88} = 23393656$, which is not zero, so $q_{\min}(0) \neq 0$ and $q_{\min}$ has no root at the origin. A solution of a constant-coefficient recurrence is a combination of terms $n^k\lambda^n$ with λ a root of its characteristic polynomial, and only the components with $\lambda = 0$ vanish after finitely many terms. Since $q_{\min}$ has none, no component of a dies out, and the minimal recurrence of every further tail is again $q_{\min}$ — the order cannot drop below 88 at any later starting point.

Now let q be the characteristic polynomial of whatever recurrence the entry asserts. The entry asserts it has order 88, so $\deg q = 88$, and it is monic. It holds from some index t on. From $\max(t, 21)$ on, the sequence satisfies both q and $q_{\min}$, and the minimal polynomial of that tail is $q_{\min}$ by the previous paragraph; therefore $q_{\min} \mid q$. Both are monic of degree 88, so $q = q_{\min}$. $\square$

5 Verification

Four checks, all in exact integer or rational arithmetic.

First, the walk model reproduces all 19 terms the entry publishes, exactly. This is what ties the model to the entry, and it is the only place where reading the entry’s English enters the argument.

Second, the order was computed twice by independent routes: once modulo a 61-bit prime, where every intermediate is a machine word, and once over $\mathbb{Q}$ in exact rational arithmetic. Both return 88, and the exact run additionally confirms every coefficient is an integer — had any been a proper fraction the recurrence would not be the entry’s.

Third, the recovered recurrence was evaluated directly on the entry’s own published terms, with no matrices and no Berlekamp–Massey involved. It holds at each of the 0 indices where the entry publishes enough earlier terms to test it.

Fourth, the last coefficient was checked to be nonzero, which is what the identification above turns on; it is -23393656 .

Remark 4. The conjecture’s own statement was never read. The entry pins it down to the family of order-88 recurrences this sequence satisfies from some point on, and that family turns out to have exactly one member.

References

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